

Vikas Murali

San Diego, CA | +1 (469)-389-7093 | vikas.murali01@gmail.com | <https://www.linkedin.com/in/vikasmurali-bioengineering/>

EDUCATION

University of California, San Diego

Expected June 2027

Jacobs School of Engineering, BS in Bioengineering: Biotechnology, Minor in Biology

Organizations: Biomedical Engineering Society, UCSD Model UN (TritonMUN 29 & 30 Chair), Sitaare A Capella (Finance)

Coursework: Genetics, Molecular Biology, Cell & Tissue Engineering, MATLAB, Fluid Mechanics, Organic & Bio Chemistry

Coursera: Genomic Data Science (Johns Hopkins), Medical Software (Yale), Biomedical Visualisation (University of Glasgow)

PROFESSIONAL EXPERIENCE

Alexandrov Lab, Undergraduate Researcher, San Diego, CA

May 2025 - Present

- Performing high-throughput DNA extraction, qPCR, and bulk/UD sequencing for somatic mutation profiling in tumor samples, automating workflows with Integra robotics and Vialab scripting to achieve 3× higher throughput and accuracy
- Processing 1,000+ TCGA breast cancer exome datasets by filtering validated mutations, generating tumor-specific variant files, managing preprocessing and computing workflows on the UCSD Supercomputing Cluster for mutational signature analysis
- Running SBS96/SBS192 signature extractions using SigProfilerExtractor with tuned NMF parameters (replicates, convergence, GMM normalization), scaling across multi-core nodes to optimize runtime and resolution of genomic profiles

Element 5, Research and Development (R&D) Intern, San Jose, CA

June 2025 - September 2025

- Analyzed 70+ Medicare audit protocols with R&D team, mapping detailed clinical specifications to structured EHR data fields (vitals, lab values, physician assessments) to build standardised, reproducible compliance criteria for patient documentation workflows
- Engineered cross-domain data validation pipelines integrating clinical notes, diagnostic reports, and longitudinal outcomes into automated audit checks, boosting data validation efficiency by approximately 28% across pilot healthcare systems
- Developed and implemented rule-based NLP-assisted workflows to flag inconsistent data within large-scale health record systems, reducing audit turnaround time and strengthening overall quality assurance processes across healthcare clients

JOGO Health, Biomedical Intern, Old Bridge, NJ

June 2024 - September 2024

- Interpreted EMG signal data and standardized clinical outcome measures (HIT-6, MSK-HQ) from digital biofeedback therapy sessions for neuromuscular disorders, identifying correlations between waveform patterns and quantifiable symptom improvement
- Authored detailed scientific case studies documenting electrode configurations, stimulation parameters, and longitudinal patient outcomes, collaborating with the legal team on intellectual property documentation for emerging digital therapeutic technologies

PROJECTS

UYIR Health Platform

March 2025 - Present

- Designing a biomedical knowledge graph (granted **Provisional Patent #63/898,118**) using LLMs to extract and link multi-source medical data (scientific studies, clinical trials, biomarkers) into an adaptive ontology enabling evidence-based therapeutic discovery
- Invented a proprietary trust-scoring algorithm (granted **Provisional Patent #63/895,716**) quantifying medical evidence credibility by combining peer-review consistency, regulatory validation, and network-based authority propagation across clinical domains
- Architecting a patent-pending AI-driven clinical decision support platform that integrates patient EHR, wearable, and genomic data with the knowledge base to generate transparent, evidence-weighted recommendations for personalized health optimization

Continuous Stress Monitoring Device

April 2023 - Present

- Ideating a wearable Continuous Stress Monitoring (CSM) device integrating electrochemical cortisol, heart rate variability, and galvanic skin response sensors for real-time stress biomarker acquisition and multi-signal fusion analysis
- Implementing embedded signal-processing and wireless telemetry systems using MATLAB, Python, and BLE microcontrollers to filter, model, and transmit biosignals for continuous cloud-based stress pattern analytics and predictive monitoring

Hashimoto's Thyroiditis Research, Chatbot

April 2022 - February 2023

- Curated and synthesized 60+ peer-reviewed studies on Hashimoto's thyroiditis to build a clinical evidence database, created an AI chatbot to support autoimmune patients with personalized, research-driven recommendations on nutrition and symptom management
- Built a transformer-based NLP pipeline using Python, spaCy, and Hugging Face models for entity extraction, intent recognition, and response ranking, allowing real-time generation of context-aware, patient-specific therapeutic guidance

ACTIVITIES

Biomedical Engineering Society, Wet Lab & Workshop Lead, San Diego, CA

October 2024 - Present

- Conducting wet lab sessions on PCR, plasmid cloning, gel electrophoresis, and ELISA assays, ensuring biosafety compliance, and mentoring 40+ undergraduates to strengthen their technical proficiency for competitive research placements
- Led hands-on workshops on cell culture, microscopy, and DNA sequencing, developing structured training materials and guiding students in experimental design, data analysis, and biomedical instrumentation to enhance technical proficiency

Ideas2IT Foundation, Volunteer, Chennai, India

September 2024 - Present

- Coordinated logistics and on-site operations for community food drives, distributing 15,000+ meals across underserved areas
- Spearheading digital fundraising and social media campaigns, boosting donor engagement by 32% and reaching 20,000+ users

Technical Skills

DNA/RNA Extraction/NGS Sequencing, RT-PCR, CRISPR, Western Blot, CAD (SolidWorks), MATLAB, Data Analysis (Python, R), Mammalian Cell Culture, Immunoassays, Quality Assurance, Presentation Decks, Microsoft Word/PowerPoint/Excel, Agile, Aseptic Technique, Automation, cGMP/GMP, SOPs, Good Documentation Practices (GDP), Lab Notebook Documentation